In [1]:
```
pip install pandas
```

```
Requirement already satisfied: pandas in C:\Users\DELL XPS\anaconda3\Lib\site-packages (3.0.3)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\DELL XPS\anaconda3\Lib\site-packages (from pandas) (2.4.
6)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\DELL XPS\anaconda3\Lib\site-packages (from pan
das) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\DELL XPS\anaconda3\Lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in C:\Users\DELL XPS\anaconda3\Lib\site-packages (from python-dateutil>=
2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import pandas as pd
data=pd.read_csv(r"C:\Users\DELL XPS\Downloads\trends.csv")
```

In [3]:
```
data
```

Out[3]:

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
...	...	...	...	...	...
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miễn thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

26955 rows × 5 columns

In [4]:
```
data.head()
```

Out[4]:

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

In [5]:
```
data.tail()
```

Out[5]:

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miễn thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

In [6]:
```
data.shape
```

Out[6]:
```
(26955, 5)
```

In [7]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   location  26955 non-null  str
 1   year      26955 non-null  int64
 2   category  26955 non-null  str
 3   rank      26955 non-null  int64
 4   query     26955 non-null  str
dtypes: int64(2), str(3)
memory usage: 2.0 MB
```

In [8]:
```
data.describe()
```

Out[8]:

	year	rank
count	26955.000000	26955.00000
mean	2015.243369	3.00000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

In [9]:
```
data.columns
```

Out[9]:
```
Index(['location', 'year', 'category', 'rank', 'query'], dtype='str')
```

In [10]:
```
data.dtypes
```

Out[10]:
```
location      str
year        int64
category      str
rank        int64
query         str
dtype: object
```

In [11]:
```
data.isnull()
```

Out[11]:

	location	year	category	rank	query
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
26950	False	False	False	False	False
26951	False	False	False	False	False
26952	False	False	False	False	False
26953	False	False	False	False	False
26954	False	False	False	False	False

26955 rows × 5 columns

In [12]:
```
data.sum
```

Out[12]:
```
<bound method DataFrame.sum of       location  year         category  rank               query
0       Global  2001  Consumer Brands     1               Nokia
1       Global  2001  Consumer Brands     2                Sony
2       Global  2001  Consumer Brands     3                 BMW
3       Global  2001  Consumer Brands     4                Palm
4       Global  2001  Consumer Brands     5               Adobe
...        ...   ...             ...   ...                 ...
26950  Vietnam  2020          Là Gì?     1   Virus Corona là gì
26951  Vietnam  2020          Là Gì?     2  Miễn thị thực là gì
26952  Vietnam  2020          Là Gì?     3    Đầu cắt moi là gì
26953  Vietnam  2020          Là Gì?     4  Bệnh bạch hầu là gì
26954  Vietnam  2020          Là Gì?     5       Đông Lào là gì

[26955 rows x 5 columns]>
```

In [13]:
```
data.duplicated()
```

Out[13]:
```
0        False
1        False
2        False
3        False
4        False
         ...
26950    False
26951    False
26952    False
26953    False
26954    False
Length: 26955, dtype: bool
```

In [14]:
```
data.sum()
```

Out[14]:
```
location    GlobalGlobalGlobalGlobalGlobalGlobalGlobalGlob...
year                                                 54320885
category    Consumer BrandsConsumer BrandsConsumer BrandsC...
rank                                                    80865
query       NokiaSonyBMWPalmAdobeNostradamusOsama bin Lade...
dtype: object
```

In [15]:
```
data.nunique()
```

Out[15]:
```
location       83
year           20
category     2450
rank            5
query       18431
dtype: int64
```

In [ ]:
```

```